MIL-E-1/17C
4 December 1968
SUPERSEDING
MIL-E-1/17B
21 January 1959

MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES OA3 AND OA3A 1/
a b

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Voltage regulator

				©
Outline	a	---	12-7 (EIA)	©
	b	---	9-7 (EIA)	
Base	a	---	B6-3	
	b	---	B6-8	
Envelope	a	---	ST12	
	b	---	T9	
Cathode	a, b	---	Glow discharge	

Base connections:

Pin No.	a, b	---	1	2	3	5	7	8
Element		---	nc	k	7 (see note 2)	a	3 (see note 2)	nc

ABSOLUTE-MAXIMUM RATINGS:

Parameter:		Ebb	Starting voltage	Operating voltage	Operating current	Alt
Unit:		Vdc	Vdc	Vdc	mAdc	ft
Maximum:	a, b	---	---	75 (approx)	40	10, 000
Minimum:	a, b	---	105	---	5	---
TEST CONDITIONS:	a, b	150	---	---	(see note 1)	---

GENERAL:

Qualification - Not required

1/ To identify those tests that are applicable to a given type or to several types; tube types are designated by letters.

© denotes changes

OA3, OA3A
a b
FSC 5960

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3347	Ionization voltage (1) (with illumination)	a, b	Rp Ib = 5 to 40 mAdc (see notes 3 and 6)	0.65	II	Ez	---	105	Vdc
3337	Voltage drop (1)	a, b	Rp Ib = 5 mAdc	0.65	II	Etd (1)	70	---	Vdc
3337	Voltage drop (3)	a, b	Rp Ib = 40 mAdc	0.65	II	Etd (3)	---	81	Vdc
3335	Regulation (1)	a, b	Etd (3) - Etd (1)	0.65	II	Reg (1)	---	6.5	Vdc
3278	Noise and oscillation (oscillation test)	a, b	Esig = 100 mVac; RL = 500 ohms; Ebb Ib = 5 to 40 mAdc (see note 4)	0.65	II	---	---	---	---
3278	Noise and oscillation (noise test)	a, b	Ebb Ib = 40 mAdc; RL = 500 ohms (see note 4)	0.65	II	Eb	---	5.0	mVac
1201	Short and discontinuity detection	a, b		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	a, b	No voltage	6.5	S3	---	---	---	---
3337	Voltage drop (2)	a, b	Rp Ib = 30 mAdc	6.5	S3	Etd (2)	---	79	Vdc
3335	Regulation (2)	a, b	Etd (2) - Etd (1)	6.5	S3	Reg (2)	---	4.5	Vdc
3305	Leakage current	a, b	Eb = 50 Vdc	6.5	S3	LIb	---	10.0	µAdc
1101	Secureness of base	a, b		6.5	S3	---	---	---	---
1111	Base pin solder depth	a, b	See note 7	6.5	S3	---	---	---	---
1105	Permanence of marking	a, b		---	---	---	---	---	---
	Quality conformance inspection, part 3								
1501	Intermittent life	a, b	Group A; Rp Ib = 30 mAdc	---	---	---	---	---	---
---	Life-test end points (intermittent) (500 hours)	a, b	Ionization voltage (see note 5)	---	---	Ez	---	105	Vdc
			Voltage drop (2)	---	---	Etd (2)	---	83	Vdc
			Voltage drop (1)	---	---	Etd (1)	68	---	Vdc
			Regulation (1) (see note 5)	---	---	Reg (1)	---	6.5	Vdc
			Regulation (2) (see note 5)	---	---	Reg (2)	---	4.5	Vdc

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NOTES:

1. Fixed resistor may be used and Ebb varied to give desired current.
2. Pins 3 and 7 tied internally.
3. Test to be conducted with tube exposed to light intensity of 5 to 50 foot candles.
4. The internal connection between pin 3 and pin 7 shall be connected in series with the cathode during this test.
5. The tube shall have reached the end of its life with regard to ionization when the ionization voltage first exceeds the value of 105.0 volts, when regulation (1) first exceeds the value of 6.5 volts, or when regulation (2) first exceeds the value of 4.5 volts.
6. This test to be performed at the conclusion of the holding period.
7. This test applies for flexible leads as well as for rigid leads.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

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